

# Subhav Kumar

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## EDUCATION

**B.Tech in Computer Science & Engineering**  
Indian Institute of Information Technology (IIIT), Raichur

2023 – 2027  
**CGPA: 8.99 / 10.00**

## EXPERIENCE

**AI Research Intern — WSAI, IIT Madras** May 2026 – Present

- Researching **Semantic Text Reconstruction from Embeddings** — investigating decoder-based methods to recover meaningful text from dense vector representations.
- Building evaluation tooling in **Python** to measure reconstruction fidelity, semantic similarity, and latent space geometry across transformer architectures.
- Running large-scale experiments on **HPC/GPU clusters**, optimizing training pipelines for throughput and memory efficiency.

## RESEARCH

**Paper Accepted — IEEE ICCSE 2026** Budapest, Hungary

- *"PCA-based Defense Against Embedding Inversion and User History Leakage in Recommendation Systems."*
- Proposed a **PCA-based dimensionality reduction** defense that mitigates embedding inversion and membership inference attacks on recommendation system user embeddings.
- Preserved recommendation quality while significantly reducing attacker reconstruction accuracy.

## PROJECTS

**Semantic Text Reconstruction from Embeddings** Ongoing

- Built a decoder-only transformer pipeline in **PyTorch** to reconstruct natural language text from sentence embeddings.
- Implemented custom training loops with attention masking, embedding injection, and beam search decoding.
- Evaluated output using BLEU, cosine similarity, and semantic equivalence metrics across multiple embedding models.
-  [View Project Repository](#)

**Student Analytics Dashboard** Aug – Sep 2025

- Developed a full-stack analytics platform using **React**, **Node.js**, **Express**, and **MongoDB** with interactive data visualizations.
- Implemented **JWT authentication** and **Role-Based Access Control (RBAC)** for secure multi-user access.
- Built RESTful APIs with input validation, error handling, and optimized database queries.
-  [View Project Repository](#)

**CCTV Face Enhancement System** Nov 2025 – Jan 2026

- Engineered a **computer vision** pipeline to detect, crop, and enhance low-resolution faces from CCTV footage using super-resolution models.
- Integrated face detection (**OpenCV**) with upscaling modules; optimized inference latency for near real-time processing.
-  [View Project Repository](#)

## TECHNICAL SKILLS

|                          |                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------|
| <b>Languages</b>         | Python, JavaScript, TypeScript, C/C++, SQL                                                    |
| <b>AI / ML</b>           | PyTorch, TensorFlow, Scikit-learn, Transformers, NLP, Computer Vision, LLM Fine-tuning (LoRA) |
| <b>Web Dev</b>           | React, Next.js, Node.js, Express, FastAPI, Django, REST APIs, HTML/CSS, TailwindCSS           |
| <b>Tools &amp; Infra</b> | Git, Docker, Linux, AWS, MongoDB, PostgreSQL, HPC/GPU Clusters                                |

## CERTIFICATIONS

Oracle Cloud Infrastructure 2025 AI Foundations Associate  
Master SQL and DBMS — Scaler

 [Verify](#)  
 [Verify](#)

## ACHIEVEMENTS

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- **Amazon ML Challenge 2025:** Ranked in **Top 1200** out of 80,000+ participants.
- **Competitive Programming:** Solved **400+ DSA problems** (LeetCode/GFG/Codeforces). 
- **Hackathons:** Walmart Sparkathon  and Adobe India Hackathon. 
- **PRMO Qualified** — JEE Mains 2023 — AIR **35,000**.